# SCHOOL OF ENGINEERING
## PROJECT WEEKLY PROGRESS REPORT

| Name : | Admin No : | Module Code : | Week No : | Date of Monday : |
|---|---|---|---|---|
| **Lee Dong Geon** | 242558W | | 4 | 30/3/2026 |

Objectives of the Week :

This week, the main objective was to continue refining and improving the CV screening system by enhancing the RAG pipeline and optimising the overall model performance. I also planned improvements for the existing code structure to make the system more stable, accurate, and easier to maintain.

Another key objective was to experiment with different RAG configurations, like chunk size, overlap, and retrieval count (k value), to identify the most effective setup that extracts relevant information from resumes to improve the accuracy of candidate evaluation and reduce wrong scoring caused by irrelevant or missing context.

The goal was also to complete the remaining enhancements planned for Week 5 ahead of schedule like improving score extraction and refining the pipeline logic,

Objectives Achieved and Work Done :

During this week, I made significant progress as both Week 4 and Week 5 tasks were completed ahead of schedule.

One of the main improvements that I made was to refine the RAG pipeline using LangChain and FAISS. Instead of using a fixed configuration, I tested multiple chunking configuration, including different chunk sizes, overlaps, and k values. I ran these experiments to see how different settings would affect how well the model can retrieve the most relevant parts of each resume.

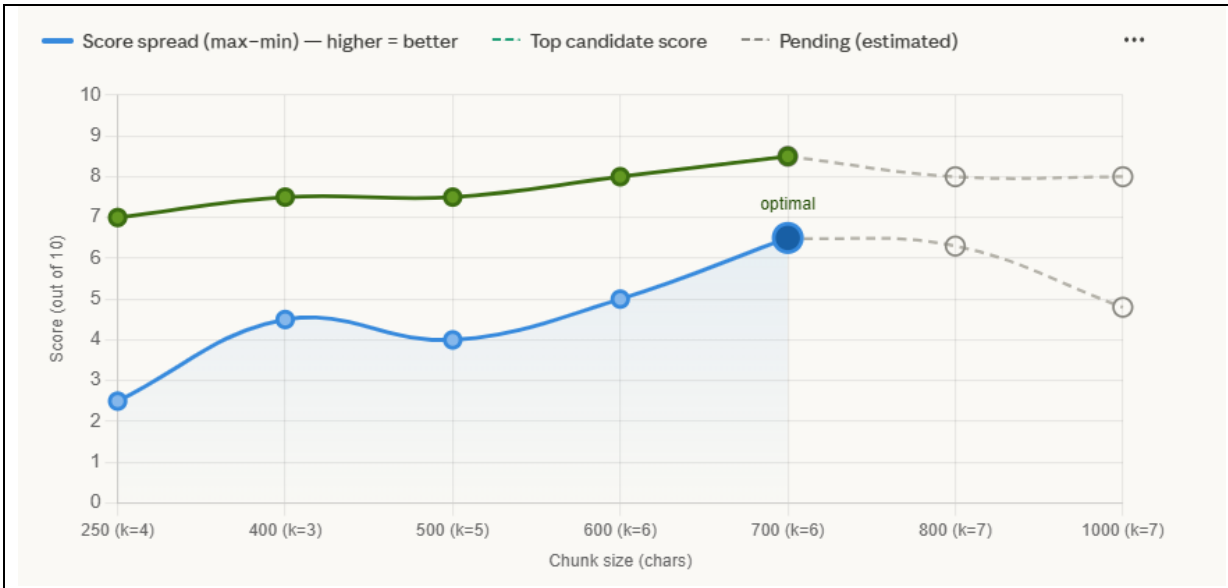
From these tests, I noticed that smaller chunk sizes (like 250 characters) tend to lose too much context, which caused the model to give very similar scores across candidates or even ties. On the other hand, larger chunk sizes were better at keeping important details like job roles and skills, but if the chunks were too big, they started to include extra unnecessary information. Based on this, I chose chunk size = 700, overlap = 175, and k = 6 because it gave the best balance between keeping enough context and staying relevant, which can also be seen from the comparison graph.

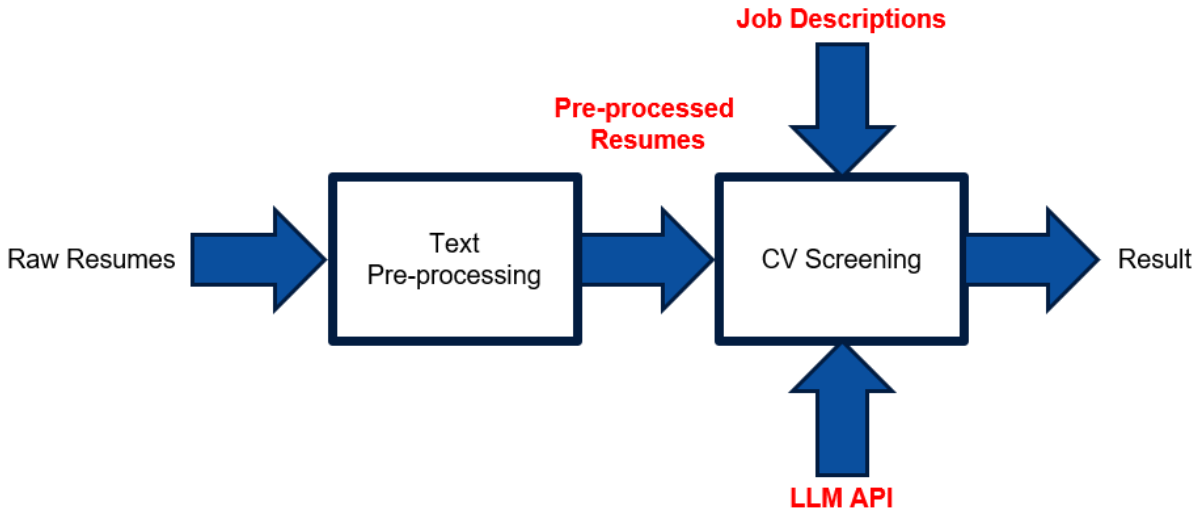
I also improved the score extraction part of the system. I used more robust pattern matching so that the scores can still be accurately extracted even when the LLM responses have different formats. This helped reduce cases where scores were missing or wrongly captured during evaluation.

I cleaned up and reorganised the code. Unnecessary or experimental code from previous weeks was removed, and related steps were grouped together into more structured sections. This made the whole pipeline easier to read, debug, and maintain moving forward.

Security and usability were also improved by changing how the API key is handled. Instead of hardcoding it directly in the code, I used a hidden input prompt to enter the API key. This makes sure sensitive information isn't exposed, which makes the system more secure and closer to real-world practice.

Finally, both the Non-RAG and RAG pipelines were fully completed and integrated. The system can now process resumes, evaluate candidates using the LLM, compare results from both approaches, and generate a final recommendation. Overall, the system is now more accurate, stable, and better optimised compared to the previous weeks.




| Task / Type | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | W11 | W12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Research LLM APIs (Target) | | | | | | | | | | | | |
| Research LLM APIs (Actual) | | | | | | | | | | | | |
| Select LLM + preprocessing (Target) | | | | | | | | | | | | |
| Select LLM + preprocessing (Actual) | | | | | | | | | | | | |
| Develop screening (LLM + RAG) (Target) | | | | | | | | | | | | |
| Develop screening (LLM + RAG) (Actual) | | | | | | | | | | | | |
| Refine screening system (Target) | | | | | | | | | | | | |
| Refine screening system (Actual) | | | | | | | | | | | | |
| Optimise performance (Target) | | | | | | | | | | | | |
| Optimise performance (Actual) | | | | | | | | | | | | |
| Mid Presentation (Target) | | | | | | | | | | | | |
| Mid Presentation (Actual) | | | | | | | | | | | | |
| Develop UI (Target) | | | | | | | | | | | | |
| Develop UI (Actual) | | | | | | | | | | | | |
| Refine UI (Target) | | | | | | | | | | | | |
| Refine UI (Actual) | | | | | | | | | | | | |
| Deployment improvements (Target) | | | | | | | | | | | | |
| Deployment improvements (Actual) | | | | | | | | | | | | |
| Enhance deployment (Target) | | | | | | | | | | | | |
| Enhance deployment (Actual) | | | | | | | | | | | | |
| Write report (Target) | | | | | | | | | | | | |
| Write report (Actual) | | | | | | | | | | | | |
| Final submission (Target) | | | | | | | | | | | | |
| Final submission (Actual) | | | | | | | | | | | | |

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| Gantt chart with updated progress. |
| Supervisor's Comment :<br><br>_____<br>Date/Signature |